

Box Model

Creating Space for Your Elements

Height and Width

- The default width of inline elements is the content.
- Elements that are not inline can take width and height properties – we saw this in the Display lecture.

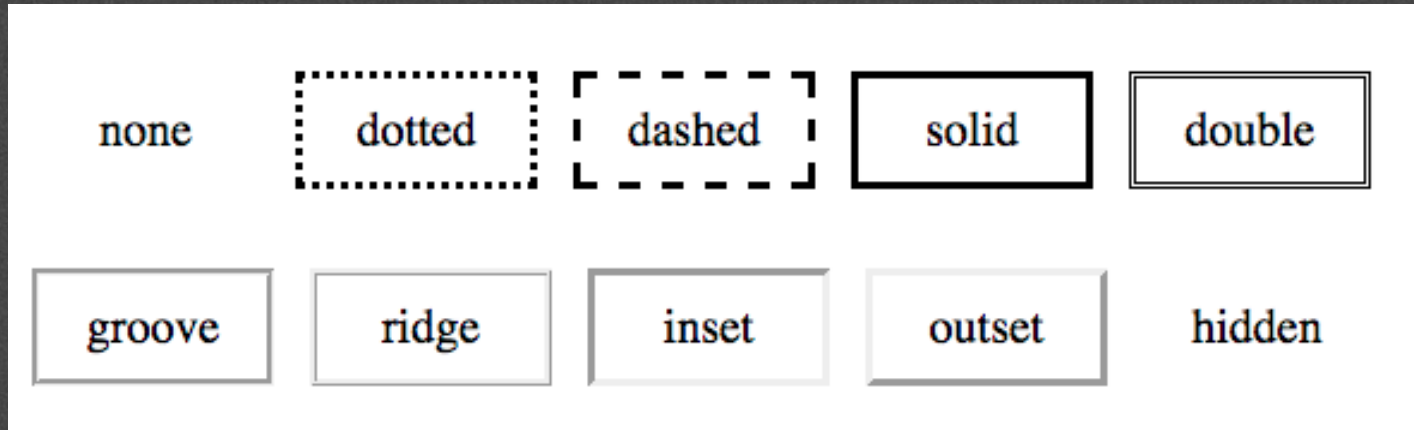
Border

- Any element can have a border around it
- border property specifies *style*, *width*, and *color*
- The border style **MUST** be specified

```
div {  
    border: solid 1px #CC00AA;  
}
```


Border-style

- none, dotted, dashed, solid, double, groove, ridge, inset, outset, hidden



Border width and color

- **Width**
 - Set in pixels or *thin, medium, or large*
- **Color**
 - **Name** - “blue”
 - **RGB** – `rgb(0,0,255)`
 - **hex** - `#0000FF`
 - **transparent**

Specifying Individual Sides

```
border-width: 3px;
```

```
border-width: 3px 10px;
```

```
border-width: 3px 10px 20px;
```

```
border-width: 3px 10px 20px 1px;
```

Borders!

Borders!

Borders!

Borders!

Margin

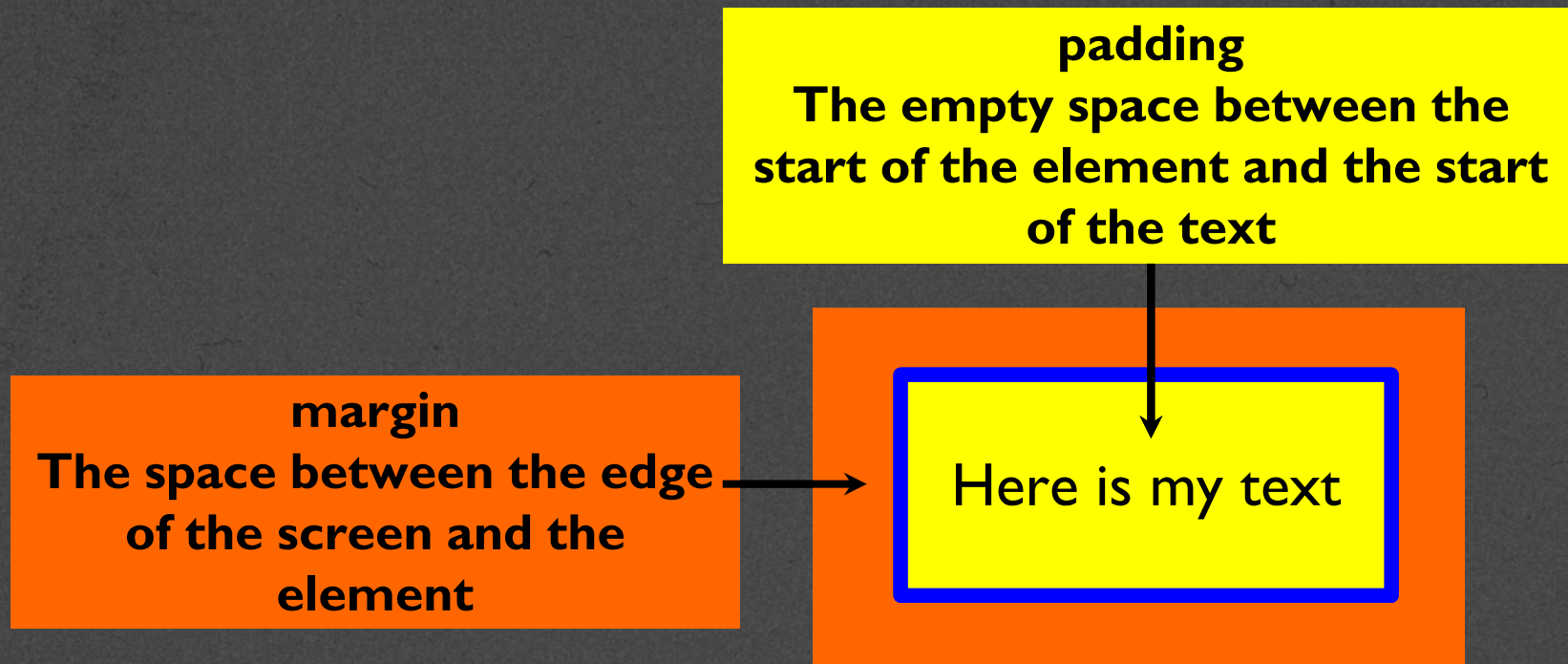
- **Margin is additional space outside your border – between you and neighbor**
- **Positive margin**
 - **element moves right/down**
- **Negative margin**
 - **element moves left/upward**

Padding

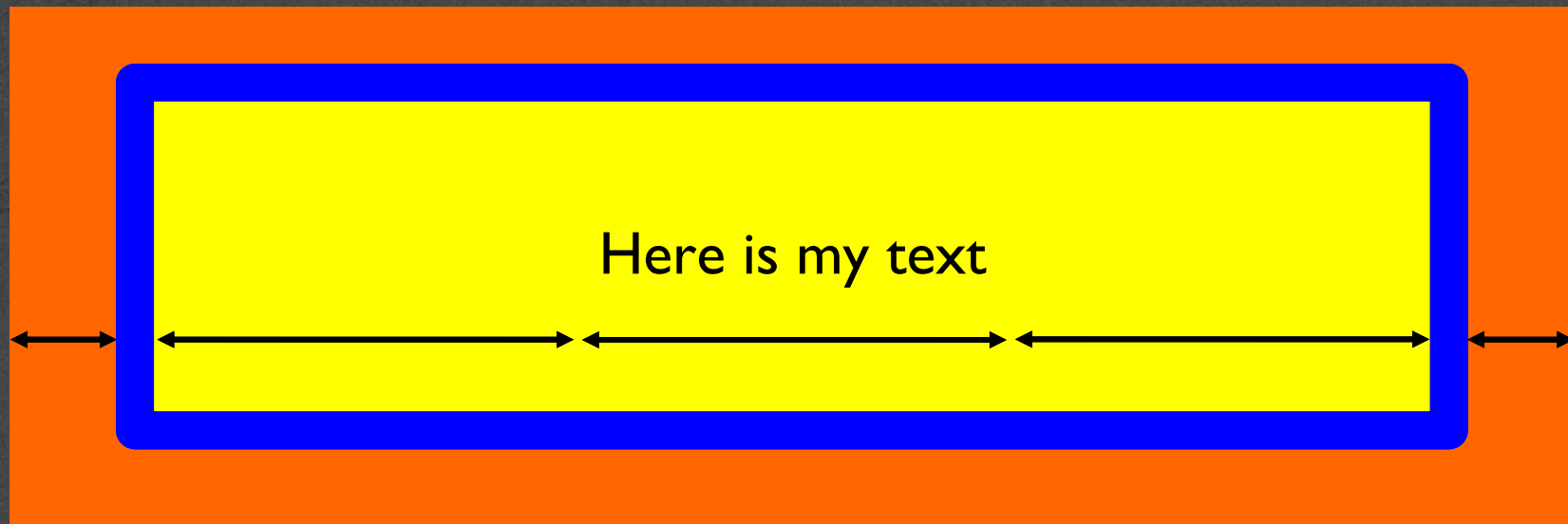
- **Padding is additional space between the element and its border.**
- **Positive padding**
 - **border moves outward from element**
- **Negative padding**
 - **border moves over the element**

Margin and Padding

- **Neither takes a color (transparent)**
- **Can also be defined in 1 - 4 values like border**



Additive Height and Width



margin + border + padding + width = actual width

What is the width and height?

```
div {  
    width: 100px;  
    height: 50px;  
    padding: 10px;  
    margin: 5px;  
    border: 1px solid black;  
}
```

Centering an Element

- To horizontally center an element use:
 - `margin: 0 auto;`
- But...
 - The element must display: block
 - The element must not float
 - *The element must not have a fixed or absolute position*
 - The element must have a width that is not auto

box-sizing

- **box-sizing** takes some of the “math” out
- **Options:**
 - **content-box:** default additive
 - **border-box:** width takes content, padding, and border into consideration

Measurements

- **Absolute – set to a specific size**
 - px, mm, cm, pt,
- **Fluid – sets size relative to surrounding elements**
 - %, vw, vh
 - em (for font): 1em is current size, .75 is 75% of the current size
 - rem (for font): 1rem is current size of root element

Review

- **Design sketches should be done with box model (margin, border, padding, content) in mind.**
- **Use box-model to reduce complexity**
- **Margin must always be considered**
- **Use fluid sizes for best viewing**

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Fluid Units and Overflow

Sizing and fitting your content

CSS Units

- **Some measurements are fixed:**
 - **font-size: 100pt**
 - **width: 500px**
- **Some measurements are fluid:**
 - **grid-template-columns: 45% 45%**
 - **height: 50vh**
 - **font-size: 2em**

Overflow

- Sometimes measurements still don't work or are out of our control.
- You can use the `overflow: property` to control what should happen

Overflow options

- **visible** - Show all of the content, even if it doesn't fit.
- **hidden** - The content that doesn't fit will be invisible.
- **scroll** - Only the content that fits is visible, and a scrollbar is added to see the rest of the content
- **auto** - Similar to scroll, but it adds scrollbars only when necessary

Example

- [Replit: Simple Grid With Overflow](#)

Accessibility Issues

- **Avoid using hidden if there is any text or important visual imagery. Just because you "think" it fits doesn't mean it will for everyone – some people magnify the page**
- **If you use scroll or auto you should add `tabindex="0"` to the html of the element you are styling so that it is still keyboard accessible.**

Visibility

- Specifies whether or not element is visible
- Options include:
 - visible
 - hidden
 - collapse (only for table elements)
- Unlike `display:none` a hidden element is still part of the DOM and still takes up space

Review

- **When setting the size, overflow and visibility of your elements, remember that not everyone accesses your site the same.**
- **Think about what parts of your site have content, vs which parts are just decorative.**

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Positioning

Positioning!

- **Putting elements where you want them can be time-consuming and frustrating**
- **Why not tables?**

Position Properties

- **The four position properties are:**
 - **static**
 - **relative**
 - **absolute**
 - **fixed**
- **Position properties are modified by the properties: top, right, bottom, left**

Static

- **Default value for elements**
- **Place in the next available position**
- **Not affected by the top, bottom, left, and right properties.**

Relative

- **Positioned “relative to itself”**
- **Take the static position, but add offsets.**
- **The new positioning does NOT affect any other element. It is possible to move an element and leave a big hole where it would have been.**
- **Relatively positioned elements are often used as container blocks for absolutely positioned elements.**

Absolute

- Element is removed from the document flow and positioned relative to its *nearest ancestor* (or the root)
- Other elements behave as if element does not exist
- Can end up on top of another element

Fixed Position

- Positioned relative to the *browser window*
- Will not move, even if the window is scrolled
 - IE7 and IE8 support the fixed value only if a **!DOCTYPE** is specified
- Think of popup boxes that won't go away!!!
- Or a navigation bar that is always visible on the top

Z-index

- **Multiple elements may be placed in the same position.**
- **z-index is a numeric value, positive or negative that dictates stacking order**

Review

- **Positioning elements is key to achieving desired layouts**
- **Proper planning will make this easier**

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